

Text-as-Data Methods in Political Science

Zach Dickson & Dan de Kadt

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GV330

Overview

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Text Analysis:
Converting letters
to numbers

Machine Learning
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Applications of
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Text as Data: an "embarrassment of riches"

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A huge amount of politics happens through **language**:

- ▶ Political speeches, legislative debates, committee transcripts, party manifestos, lobbying documents, campaign websites
- ▶ Bills, laws, treaties, constitutions, court opinions and judgments
- ▶ News articles, opinion columns
- ▶ Open-ended survey questions, interviews, focus groups
- ▶ Twitter, Facebook, Reddit, blogs, other social media

Text as data is the process of converting unstructured text into structured data that can be analyzed with statistical methods.

What could we do with all this data?

- ▶ **classify** documents as belonging to particular types
- ▶ **describe** the content and sentiment of text
- ▶ use text to **inform our coding** of variables
- ▶ **predict** outcomes based on text
- ▶ **summarize** large amounts of text
- ▶ ... and more!

Note: This is an active area of research → new methods developed all the time

From Documents to Data

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The data begin as a structured **document** (a tweet, a bill, a speech).



Economic Freedom Fighters ✓
@EFFSouthAfrica



#EFFAfricaDay DSG @hlengiwe44maxon wishing all Africans an Africa Day that celebrates the resilience and tenacity of African women. #AfricaDay

Multiple documents make a **corpus** (a collection of docs):
Naively, this is just a dataset with a character column of
“text”

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Generations of Text Analysis

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- ▶ Symbolic/rules-based methods
 - ▶ 1950s–2000s
- ▶ Bag of words (e.g. **term-document matrix**)
 - ▶ 1990s–2010s
- ▶ Embeddings (e.g. **word2vec**)
 - ▶ 2013–2018
- ▶ Transformers (e.g. **BERT, GPT**)
 - ▶ 2018–Present
- ▶ Retrieval-Augmented Generation (RAG)?

Bag of words

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Bag of words (e.g. **term-document matrix**)

- ▶ Simple, but effective
- ▶ Good for many tasks
- ▶ Easy to understand

I love this movie! It's sweet,
but with satirical humor. The
dialogue is great and the
adventure scenes are fun...
It manages to be whimsical
and romantic while laughing
at the conventions of the
fairy tale genre. I would
recommend it to just about
everyone. I've seen it several
times, and I'm always happy
to see it again whenever I
have a friend who hasn't
seen it yet!



it	6
I	5
the	4
to	3
and	3
seen	2
yet	1
would	1
whimsical	1
times	1
sweet	1
satirical	1
adventure	1
genre	1
fairy	1
humor	1
have	1
great	1
...	...

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The Corpus and Term-Document Matrix

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Converting a document into a (cleaned, sparse)
term-document matrix:

```
> tdm = TermDocumentMatrix(data_text, control = list(verbose = TRUE,
+           asplain = TRUE,
+           stopwords = TRUE,
+           tolower = TRUE,
+           removenumbers = TRUE,
+           stemwords = TRUE,
+           removepunctuation = TRUE,
+           remove separators = TRUE,
+           remove twitter = TRUE,
+           stem = TRUE,
+           stripwhitespace = TRUE,
+           wordlengths=c(3, inf)))
> m = as.matrix(tdm)
> head(m)
```

	Docs																																															
Terms	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48
bed	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
corp	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
get	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
good	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	Docs																																														
Terms	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93		
bed	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
corp	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
get	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
good	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Think of as a **bag of words/n-grams** – mostly ignore structure.

Embeddings

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- ▶ Coding words as vectors
- ▶ Captures semantic meaning
- ▶ More complex than bag of words
- ▶ Examples:
 - ▶ Word2Vec
 - ▶ GloVe
 - ▶ FastText

Word2Vec

Text as Data

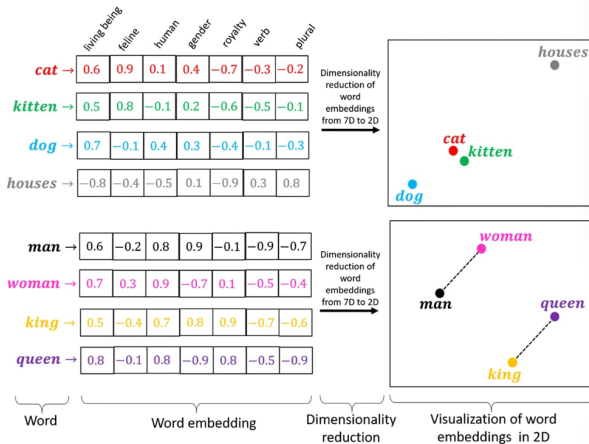
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Embeddings Example: Word2Vec (Mikolov et al. 2013)

- ▶ Goal: capture semantic meaning of different words
- ▶ Code words as vectors in a high-dimensional space
- ▶ Words that are similar in meaning are close in this space
- ▶ We can perform math operations on vectors to capture relationships

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Transformers (Vaswani et al. 2017)

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Attention Is All You Need

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.0 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature.

Transformers

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- ▶ Transformers (e.g. **BERT, GPT**)
- ▶ More complex than embeddings
- ▶ Bidirectional
- ▶ State of the art for many tasks
- ▶ Examples:
 - ▶ BERT (Bidirectional Encoder Representations from **Transformers**)
 - ▶ GPT (Generative Pre-trained **Transformers**)

Transformers Architecture

Text as Data

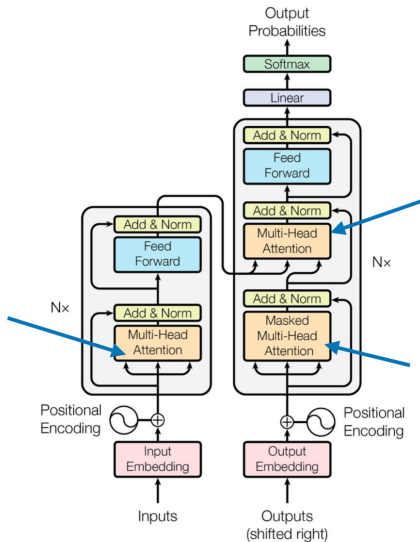
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Attention Mechanism in Transformers

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The FBI is chasing a criminal on the run .

The FBI is chasing a criminal on the run .

The FBI is chasing a criminal on the run .

The FBI is chasing a criminal on the run .

The FBI is chasing a criminal on the run .

The FBI is chasing a criminal on the run .

The FBI is chasing a criminal on the run .

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- ▶ Where does machine learning come in?
 - ▶ Dimensionality reduction (e.g. **topic modeling**)
 - ▶ Classification (e.g. **sentiment analysis**)
 - ▶ Clustering (e.g. **unsupervised learning**)
- ▶ Deep learning (e.g. **neural networks**)
 - ▶ Transformers (e.g. **BERT**)
 - ▶ Language models (e.g. **GPT**)
- ▶ Many other [statistical] methods

Supervised learning

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When we have labeled data, we can use supervised learning methods to predict outcomes based on text.

- ▶ Classification
 - ▶ Sentiment analysis
 - ▶ Topic classification
- ▶ Regression
 - ▶ Predicting a continuous outcomes based on text
 - ▶ e.g. predicting stock prices based on news articles

Unsupervised learning

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When we don't have labeled data, we can use unsupervised learning methods to explore the structure of text.

- ▶ Clustering
 - ▶ Grouping similar documents together
 - ▶ e.g. identifying different topics in a corpus
- ▶ Dimensionality reduction
 - ▶ Topic modeling
 - ▶ e.g. LDA, STM

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Example: Dickson & Hobolt (2024)

Going Against the Grain: Climate Change as a Wedge Issue for the Radical Right

Comparative Political Studies

2024, Vol. 0(0) 1–27

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Zachary P. Dickson¹ and Sara B. Hobolt¹ 

Abstract

Political parties often mobilise issues that can improve their electoral fortunes by splitting existing coalitions. We argue that by adopting a distinctively adversarial stance, radical right-wing parties have increasingly politicised climate change policies as a wedge issue. This strategy challenges the mainstream party consensus and seeks to mobilise voter concerns over green initiatives. Relying on state-of-the-art multilingual large language models, we empirically examine nearly half a million press releases from 76 political parties across nine European democracies to support this argument. Our findings demonstrate that the radical right's oppositional climate policy rhetoric diverges significantly from the mainstream consensus. Survey data further reveal climate policy scepticism among voters across the political spectrum, highlighting the mobilising potential of climate policies as a wedge issue. This research advances our understanding of issue competition and the politicisation of climate change.

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Climate Change as a Wedge Issue for the Radical Right

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- ▶ Challenger parties – such as right-wing populist or Green parties – often wish to divide existing coalitions
- ▶ Increasingly, climate change is a wedge issue that can be used to divide the traditional left-right divide
- ▶ We can use a range of text analysis methods to understand how parties use climate change as a wedge issue

Climate Change as a Wedge Issue for the Radical Right

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Wedge Issue Politics

- ▶ Argument 1: Radical right parties have increasing emphasized climate change in their communication
- ▶ Argument 2: Radical right parties take a different stance on climate change than nearly all other parties
- ▶ Argument 3: There is a significant proportion of voters who disagree with their party on climate change

Data Collection & Methods

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- ▶ Party Press releases from 2010-2023
- ▶ Press releases from 11 countries (76 parties)
- ▶ Nearly 600,000 press releases
- ▶ Language models to translate and summarize press releases
- ▶ Supervised machine learning to classify press releases

Press Release Example: UKIP (2023)

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UKIP
THE NEW RIGHT

[f](#)[t](#)[X](#)[@](#)[v](#)[y](#)

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Energy Security & Net Zero

Antony Nailer • 13 February 2023

News & Social Media / Post

Search the news

Antony Nailer
Energy Spokesman for the UK Independence Party

Grant Shapps has been in government quite a long time and as well as being Deputy Chairman of the Conservative Party he has also had the following appointments.

1. Minister for state for International Development, May 2015 till November 2015. This department finds ways to throw taxpayers money at foreign governments.
2. Secretary of State for Transport, July 2019 till September 2022. He used the Covid period to have towns in his constituency and elsewhere block off through roads with planters making them into cul-de-sacs, to continue the war against the motorist.

The justification was to encourage people to walk or cycle or use public transport to reduce pollution from car use and ultimately save the planet. His action has just speeded up the use of on-line shopping the decline of the high street.

3. Secretary of State for the Home office under Liz Truss 19-25 October 2022.
4. Secretary of State for Business, Energy, and Industrial Strategy, (BEIS).

This had an emphasis on building more wind farms which continued the decline in granting licences and planning permissions for mining of coal, drilling for oil, and included a moratorium for fracking for gas. Now we have to import these things at enormous expense, if they are available, and with the Russian gas turned off we will need to import enormous amounts of Liquefied Natural Gas from the USA to prevent gas rationing.

5. Rishi Sunak has done a mini reshuffle, a bit like tidying the deckchairs on Titanic after it hit the iceberg, and he has now put the fox into the chicken run. Grant Shapps has become Secretary of State for Energy Security and Net Zero. This is a classic oxymoron, where words have conflicting meaning, such as 'Army Intelligence'.

Recent National News

Enough is Enough

Ever wondered what makes UKIP different?

The Father of BREXIT Dies

They are certainly 'Changqing Britain'

The Power of Labour Suggestion

Reform isn't Working (even with its closest

General Election 2024 - Our Mission

UKIP & Reform UK - What's the Difference?

The bill for PP is Unustainable

Starmer's Energy Security Policy

Show More

Climate Change as a Wedge Issue for the Radical Right

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Argument 1: Radical right parties have increasing emphasized climate change in their communication

- ▶ We can use text analysis to measure the salience of climate change in party communication
- ▶ Use supervised machine learning and language models to classify press releases
- ▶ Aggregate the data to measure the salience of climate change over time
- ▶ Compare press releases from radical right parties to other parties

Salience of Climate Change in Party Communication

Text as Data

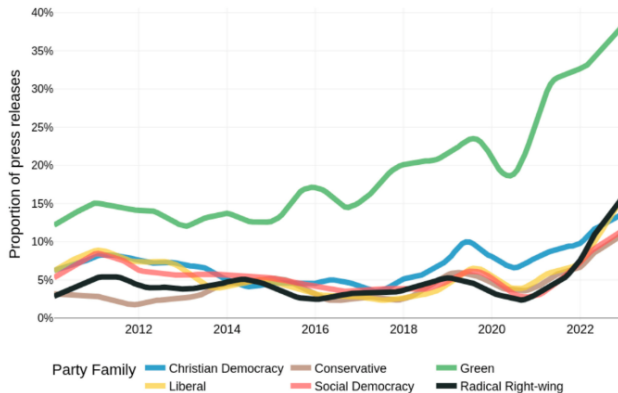
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Argument 2: Radical right parties take a different stance on climate change than nearly all other parties

- ▶ We can use text analysis to measure the stance of parties on climate change
- ▶ Train and validate a summarization model to summarize the text's position
- ▶ Aggregate the data to measure the stance of parties on climate change
- ▶ Compare the stance of radical right parties to other parties

Stance of Parties on Climate Change

Text as Data

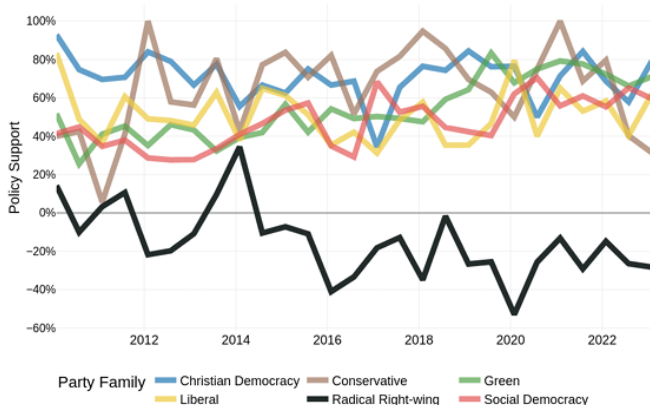
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Example: Political violence & party formation

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de Kadt, D., Johnson-Kanu, A., & Sands, M. L. (2023). Electoral accountability and state violence: The political legacy of the Marikana massacre. *American Political Science Review*.

Example: Political violence & party formation

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Example: Political violence & party formation

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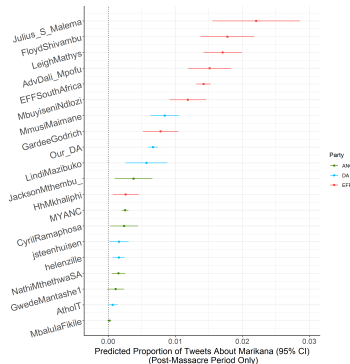
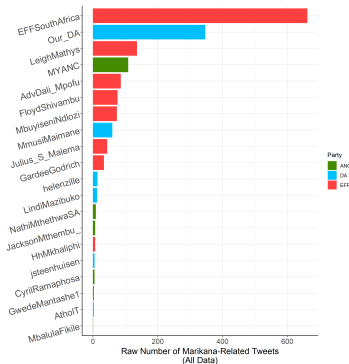
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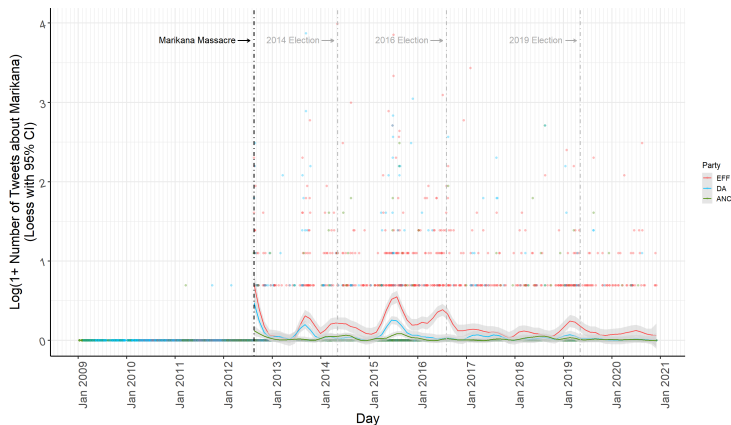
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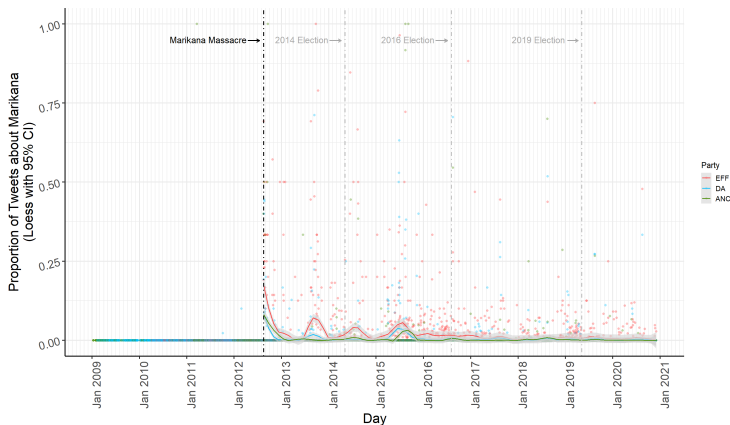
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A Simple Example: Measuring Elites' Issue Attention

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Article

The Gender Gap in Elite-Voter Responsiveness Online

Zachary P. Dickson

A number of important studies have documented gender gaps in the effectiveness or performance of individual representatives. Yet whether these differences are observable when it comes to responsiveness to public opinion is unclear. In this article, I examine the degree to which representatives use social media to dynamically respond to shifts in issue salience among the electorate. After combining nearly 400 bi-weekly repeated public opinion surveys from YouGov asking voters about their issue priorities, I trained a large language model to classify the universe of elected U.S. and UK representatives' social media messages on Twitter to the same issues. Findings reveal that women representatives demonstrate greater responsiveness than their male counterparts to shifts in issue salience according to both women and men constituents. Despite an overall bias toward male constituents, female representatives play a crucial role in narrowing the gender gap by consistently aligning their attention with the issues prioritized by female constituents. These findings not only contribute to our understanding of elite-voter responsiveness but also underscore the substantive benefits that women representatives provide for *all* constituents.

Measuring Elites' Issue Attention (Dickson (2024))

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- ▶ Do political elites respond to public opinion?
- ▶ Combine social media data, text analysis, and survey data to measure elites' responsiveness to public opinion
- ▶ Classify tweets as being about different issues using BERT transformers model
- ▶ Aggregate data to match elites' issue attention and public opinion
- ▶ Find that women are more responsive to public opinion on social media

Measuring Elites' Issue Attention (Dickson (2024))

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Figure 3
Dynamic issue attention of United States representatives by gender

